



Nikolaos -Stefanos Kostagiolas

Curriculum Vitae

Education

- 2017–2018 **M.Sc. in Data Science**, *Athens University of Economics and Business*, Greece, 8.57/10.
Awarded with excellence
- 2011–2017 **B.Sc. Informatics and Telecommunications**, *Dept. of Informatics and Telecommunications, National and Kapodistrian University of Athens*, Greece, 7.61/10.
Specialization : Data and Knowledge Management
- 2008–2011 **Apolytirion**, *2nd High School of Moschato*, Athens, Greece, 18.3/20.
Awarded with excellence

M.Sc. thesis

- title *Learning optimization algorithms with neural networks*
- supervisor Prof. Evangelos Markakis
- description Extended previous studies on meta-learned optimizers by inspecting the generalization capabilities on a variety of different settings. Awarded with the EFG Eurobank scholarship for the best MSc thesis in the program.

B.Sc. thesis

- title *A study on chromatin structure via nucleosome positioning pattern classification*
- supervisors Prof. Panagiotis Stamatopoulos, Dr. George Giannakopoulos
- description Applied text-mining on genomic sequence textual data using HMM, Bag-of-Words and N-gram Graph representations. Outperformed then state-of-the-art approaches in related studies.

Experience

Vocational

- October 2020–today **Research Assistant**, *The Cyprus Institute / National & Kapodistrian University of Athens*, Athens, Greece.
- Developed generative models for high-dimensional satellite imagery aimed at minimizing human supervision.
 - Built and released an [open-source framework](#) (PyTorch) enabling satellite image editing via latent space exploration.
 - Authored [three research papers](#) (two published in peer-reviewed venues/journals, one under review).
- February 2026 **IT & Administrative Assistant (Conscription Service)**, *Hellenic Army (Mandatory Service)*, Artillery Training Centre, Thebes, Greece.
- Handled administrative processing of military personnel transfers to other units.
 - Prepared and dispatched documentation to destination units.
 - Provided IT support and troubleshooting for office staff.
- September 2019–June 2020 **Research Assistant**, *DMML Group*, University of Geneva - HES-SO Geneve, Switzerland.
- October 2018–July 2019 **Machine Learning Engineer / Research Assistant**, *Athena Research and Innovation Center*, Athens, Greece.
- Developed open-source ML tools for geospatial data analysis using scikit-learn and TensorFlow.
 - Designed machine learning pipelines for large-scale geospatial datasets.
 - Co-authored two [research papers](#) accepted at peer-reviewed conferences.
- March 2015–October 2015 **Internship - Data Warehousing and Data Mining**, *IBM Hellas*, Athens, Greece.
- October 2015 ETL Expert contributing to the projects "Solvency II" at Ethniki Hellenic General Insurance Co. S.A. and "Energy Trading & Risk Management" for the Greek Department of Energy Management.
- Received an "Excellent" performance evaluation from supervisor Minas Koulisianis for both projects.
 - Designed and implemented the full ETL pipeline for the "Energy Trading & Risk Management" project.

Academic

- September 2015–January 2019 **Volunteer Research Associate**, *SKEL Lab - NCSR Demokritos*, Athens, Greece.
- Research in Natural Language Processing and Bioinformatics under the supervision of Prof. Panagiotis Stamatopoulos and Dr. George Giannakopoulos.
- Extended my B.Sc. thesis on nucleosome positioning pattern classification toward genomic text mining.
 - Implemented and [open-sourced](#) the proposed methods in Java.

Miscellaneous

- January 2013–June 2015 **Volunteer Teaching Assistant**, *Dept. of Informatics and Telecommunications - National and Kapodistrian University of Athens*, Athens, Greece.
- 2015 Voluntarily worked as a Teaching Assistant in Introduction to Programming and Data Structures Lab.

Publications and Preprints

1. Nikos Kostagiolas, Pantelis Georgiades, Yannis Panagakis, and Mihalís A. Nicolaou. Environment-aware satellite image generation with diffusion models, 2025
2. Georgia Kourmouli, Nikos Kostagiolas, Mihalís A. Nicolaou, and Yannis Panagakis. Locality-preserving directions for interpreting the latent space of satellite image gans. *IEEE Geoscience and Remote Sensing Letters*, 21:1–5, 2024
3. Nikos Kostagiolas, Mihalís Nicolaou, and Yannis Panagakis. Unsupervised discovery of semantic concepts in satellite imagery with style-based wavelet-driven generative models. In *Proceedings of the 12th Hellenic Conference on Artificial Intelligence*, SETN '22, New York, NY, USA, 2022. Association for Computing Machinery
4. Giorgos Giannopoulos, Konstantinos Alexis, Nikos Kostagiolas, and Dimitrios Skoutas. Classifying points of interest with minimum metadata. In *Proceedings of the 3rd ACM SIGSPATIAL International Workshop on Location-Based Recommendations, Geosocial Networks and Geoadvertising*, LocalRec '19, New York, NY, USA, 2019. Association for Computing Machinery
5. Giorgos Eftaxias, Nontas Tsakonas, Giorgos Giannopoulos, Nikos Kostagiolas, Andreas Syngros, and Dimitrios Skoutas. Lgm-pc: A tool for poi classification on qgis. In *Proceedings of the 16th International Symposium on Spatial and Temporal Databases*, SSTD '19, page 182–185, New York, NY, USA, 2019. Association for Computing Machinery
6. Nikos Kostagiolas, Nikiforos Pittaras, Christoforos Nikolaou, and George Giannakopoulos. Exploring different sequence representations and classification methods for the prediction of nucleosome positioning. *bioRxiv*, 2018

Languages

Greek	Independent user	<i>Native</i>
English	Independent user	<i>Certificate of Proficiency - Michigan, IELTS 8 Band Score</i>
French	Upper Intermediate user	<i>Delf B1</i>

Computer Science skills

Programming Languages	C++, C, JAVA, SQL, Prolog, Assembly(MIPS)	Scripting Languages	R, Matlab, Python, Unix Shell (Bash)
Machine Learning Frameworks and Toolkits	JAVA: Weka, Python: Keras, scikit-learn, PyTorch	OSs Familiarity	Windows, Linux, OS X

Research Interests

Deep Learning	Theoretical aspects of Deep Learning: Representation/Manifold Learning, Generative models for high-dimensional data, Robust and efficient methods for high-dimensional signal denoising/retrieval, Reinforcement Learning, Meta-Learning
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Multi-modal Machine Learning	Synergies between vision and language models, multi-modal semantic extraction, general purpose architectures
Machine Learning for Health Studies	Genomic text-mining, Biomedical Imaging

Open-source Code

- **SatConcepts**. Official implementation of our SETN 2022 paper "Unsupervised Discovery of Semantic Concepts in Satellite Imagery with Style-based Wavelet-driven Generative Models". Includes a framework for satellite image editing via exploration within a generative model's trained latent space. [Code on Github](#)
- **LGM-Classification**. LGM-Classification is a Python library implementing a full Machine Learning workflow for training classification algorithms on annotated POI datasets and producing models for the accurate classification of Points of Interest (POIs) into categories. [Code on Github](#)
- **LGM-Geocoding**. LGM-Geocoding is a Python library that implements a full Machine Learning workflow for training classification algorithms on annotated datasets that contain mappings between coordinate pairs and the ideal geocoding source for them, and producing models for providing accurate predictions about ideal geocoding sources for previously unseen geocoding pairs. [Code on Github](#)
- **LGM-PolygonClassification**. LGM-PolygonClassification is a Python library that implements a full Machine Learning workflow for training classification algorithms on annotated datasets that contain pairs of matched polygons each of which belongs to a distinct polygon variant. [Code on Github](#)
- **Nucleosome Positioning Pattern Recognition**. Genomic text-mining JAVA application. Developed for my B.Sc. thesis using *Weka Data Mining Software*, the *Jahmm* Library and *JInsect* Toolkit. [Code on Github](#)

Honors & Awards

- Selected among 2000 individuals to participate in the very first Climate Change AI Summer School 2022 (Acceptance Rate: 0.25%).
- Scholarship for Undergraduate Internships (Full financial support from the European Union).
- 1st position (National) and 1st position (European) in Night of the Living Devs hackathon (Organized by Microsoft Co.).
- EFG Scholarship for the best MSc thesis, MSc in Data Science, AUEB, cohort of 2017-2018.

References

Both academic and vocational references are available upon request.